Lab 08

Composition

Objectives: At the end of this lab session, the student should be able to:

- To introduce students to the concept of Java Composition and its practical applications in software development.
- Prerequisites: Basic understanding of Java programming language, classes, and objects.

Part 1: Problem Solving Exercises (Time: 60 minutes)

A class can have references to objects of other classes as members. This is called composition and is sometimes referred to as a **has-a** relationship. For example, a Customer object needs to record the Address of the client. The address is not a simple string, it is another object with many attributes.

1. Add the **Customer** class which is given below to your project.

```
public class Customer {
  private String name;
  private String phone;
  private int id;
  public Customer(String name, String phone, int id) {
    this.name = name;
    this.phone = phone;
    this.id = id;
  }
  public String getName() {
    return name;
  }
  public void setName(String name) {
    this.name = name;
  }
  public String getPhone() {
    return phone;
  }
  public void setPhone(String phone) {
    this.phone = phone;
  }
  public int getId() {
    return id;
  }
  public void setId(int id) {
    this.id = id;
  }
  public String toString() {
    return "Name=" + name + ", phone=" + phone + ", id=" + id ;
  }
```

```
}
```

2. Add the **Address** class which is given below to your project.

```
public class Address {
	private String street;
	private String city;
	private String state;
	private String zip;

	public Address(String street, String city, String state, String zip) {
		this.street = street;
		this.city = city;
		this.state = state;
		this.zip = zip;
	}
	public String getStreet() {
		return street;
	}
	public String getCity() {
		return city;
	}
	public String getState() {
		return state;
	}
	public String getZip() {
		return zip;
	}

	public void setStreet(String street) {
		this.street = street;
	}

	public void setCity(String city) {
		this.city = city;
	}
	public void setState(String state) {
		this.state = state;
	}

	public void setZip(String zip) {
		this.zip = zip;
	}
	@Override
	public String toString() {
		return street + ", " + city + ", " + state + " " + zip;
	}
}
```

3. Modify class **Customer** by adding new data member called address of type **Address**.
4. Add a fully parameterized constructor in the Customer class that initializes all attributes including the address.
5. Add setter and getter for the address data member.
6. Modify the toStirng() method by adding the address toString to the returned string.
7. Create two Customer objects and set their addresses.

8. Test your code.

Part 2: Problem Solving Exercises (Time: 60 minutes)

1. Consider the following **Book** and **Author** classes. Add them to your project.

```
public class Book {
  private String isbn;
  private String title;
  private double price;
  public Book(String isbn, String title, double price) {
    this.isbn = isbn;
    this.title = title;
    this.price = price;
  }
  public String getIsbn() {
    return isbn;
  }
  public void setIsbn(String isbn) {
    this.isbn = isbn;
  }
  public String getTitle() {
    return title;
  }
  public void setTitle(String title) {
    this.title = title;
  }
  public double getPrice() {
    return price;
  }
  public void setPrice(double price) {
    this.price = price;
  }
  public String toString() {
    return "ISBN:" + isbn + ", Title:" + title + ", Price=" + price ;
  }
}

public class Author {
  private int id;
  private String firstName;
  private String lastName;
  public Author(int id, String firstName, String lastName) {
    this.id = id;
    this.firstName = firstName;
    this.lastName = lastName;
  }
  public int getId() {
    return id;
  }
  public void setId(int id) {
    this.id = id;
  }
 public String getFirstName() {
    return firstName;
```

```
  }
  public void setFirstName(String firstName) {
    this.firstName = firstName;
  }
  public String getLastName() {
    return lastName;
  }
  public void setLastName(String lastName) {
    this.lastName = lastName;
  }
  public String toString() {
    return "ID:" + id + ", Name: " + firstName + ", " + lastName ;
  }
}
```

2. Do the following:
    a. Make the Author as a new data member of the Book class.
    b. Add a new constructor.
    c. Create setter and getter.
    d. Add Author toString to the Book's toString.
    e. Test your code.
3. Consider the following **Publisher** class. Add it to your project

```
public class Publisher {
  private int id;
  private String name;
  private String email;
  public Publisher(int id, String name, String email) {
    this.id = id;
    this.name = name;
    this.email = email;
  }
  public int getId() {
    return id;
  }
  public void setId(int id) {
    this.id = id;
  }
  public String getName() {
    return name;
  }
  public void setName(String name) {
    this.name = name;
  }
  public String getEmail() {
    return email;
  }
  public void setEmail(String email) {
    this.email = email;
  }
  public String toString() {
    return "Publisher ID: " + id + ", " + name + ", email=" + email  ;
  }
}
```

4. Do the following:

A. Make the **Address** as a new data member of the **Publisher** class.
B. Add a new constructor.
C. Create setter and getter.
D. Add **Address** toString to the Publisher's toString.
E. Test your code.

5. Modify the **Book** class by adding the **Publisher** as a new data member. Do all the things you did before with other classes.
6. Test your code with proper data.

Part 3: Problem Solving Exercises ArrayList (Time: 30 minutes)

The Book generally contains multiple chapters. Each chapter has a name and number of pages. We will make this change to the Book class.

This time we are NOT going to add Chapter as a data member, what we need is an ArrayList of Chapter objects.

7. Consider the following Chapter class. Add it to your project.

```
public class Chapter {
  private String title;
  private int pageCount;
  public Chapter(String title, int pageCount) {
    this.title = title;
    this.pageCount = pageCount;
  }
  public String getTitle() {
    return title;
  }
  public int getPageCount() {
    return pageCount;
  }
  public void setTitle(String title) {
    this.title = title;
  }
  public void setPageCount(int pageCount) {
    this.pageCount = pageCount;
  }
  @Override
  public String toString() {
    return title + " (" + pageCount + " pages)";
  }
}
```

8. Do the following changes to the Book class:
   A. Add an ArrayList of Chapters as a new data member. No need to create or modify the constructor.
   B. Add setter and getter.
   C. Add two methods: *void addChater(Chapte c)* and void *removeChapter(Chapter c)*. The add and remove methods will make the changes directly to the ArrayList of Chapters.
   D. Add a two methods: *int getPageCount()*. This method return the total number of pages in all the chapters and *int getChapterCount()*.

E. Modify the toString method of the Book class: add the number of chapters and the total number of pages to it.

9. Test your code with proper data. Create a Book object that has three chapters, a publisher who has an address, and an author.

```
Chapter ch1 = new Chapter("Chapter 1: Introduction", 10);
Chapter ch2 = new Chapter("Chapter 2: Background", 15);
Chapter ch3 = new Chapter("Chapter 3: Results", 20);
Author author1 = new Author(1, "John", "Doe");
Publisher publisher = new Publisher(1, "Pearson", "contact@pearson.com");
Address address = new Address("123 Main St", "Anytown", "Anystate", "12345");
publisher.setAddress(address);
Book book1 = new Book("123-455-6656", "Art of War", 45.9);
book1.setPublisher(publisher);
book1.setAuthor(author1);
book1.addChapter(ch1);
book1.addChapter(ch2);
book1.addChapter(ch3);
System.out.println(book1);
```

Your output should be similar to the following:

```
Art of War[123-455-6656] by Doe, John
Published by: Pearson-contact@pearson.com, 123 Main St, Anytown, Anystate 12345
(3 chapters, 45 pages, $45.90, ISBN 123-455-6656)
```

10. FINALLY: Books can be created by more than one author! This requires you to change the Author data member to an ArrayList of Authors!

You also need to add methods like *int getAuthorCount(), void addAuthor(Author a),* void *removeAuthor(Auhor a).* Make sure you do not add duplicate author ids.
Furthermore, you need to print the list of all authors in the toString() of the Book.